MC210401424: FARIHA FATIMA

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 3 of 10 (Start time: 12:43:20 PM, 25 January 2023)

The covariant differentiation of a tensor with constant components is zero.

Select the correct option

- False
- True

Question # 4 of 10 (Start time: 12:43:56 PM, 25 January 2023)

In generalized curvilinear coordinates the divergence of vector point function $A^i = A^i(x^i)$ is defined as_____

Select the correct option

- $div(A^i) = A^i_{,i} + \Gamma^i_{ik}$
- $div(A^i) = A_{i,i} + \Gamma^i_{ik} A^k$
- $div(A^i) = A^i_{,i} + \Gamma^k_{ii} A^k$
- $div(A^i) = A^i_{,i} + \Gamma^i_{ik} A^k$

MTH623 - Tensor Analysis and its Applications (Quiz # 3) Quiz Start Time: 12:42

Question # 1 of 10 (Start time: 12:42:19 PM, 25 January 2023)

The ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____.

Select the correct option

Relo

- none of these
- quadratic transformations
- non-linear transformations
- linear transformations

correct

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 10 of 10 (Start time: 12:50:44 PM, 25 January 2023)

The covariant differentiation of a scalar is equal to _____.

Select the correct option

- scalar field itself

correct

- partial derivative of scalar field

210411522: ANEESA BATOOL

4623 - Tensor Analysis and its Applications (Quiz # 3) Quiz Start Time: 01:19 P

stion # 2 of 10 (Start time: 01:19:40 PM, 25 January 2023)

he ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____

ct the correct option

Relo

- linear transformations
- quadratic transformations
- non-linear transformations
- none of these

Click to Save Answers & Next

Question # 10 of 10 (Start time: 01:21:17 PM, 25 January 2023)

The Christoffel symbols of first and second kind are equal in any coordinate system.

Select the correct option

○	True
○	False

BC210411322: ANEESA BATOOL

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 10 of 10 (Start time: 01:21:17 PM, 25 January 2023)

The Christoffel symbols of first and second kind are equal in any coordinate system.

Select the correct option

- (●) True
- ( ) False

Question # 2 of 10 (Start time: 12:42:53 PM, 25 January 2023)

In the notation Γ_{ij}^{k} of Christoffel symbol of second, the symbol k tells_____

Select the correct option

- ○ none of these
- ○ which basis vector points in the direction of this component of the derivative vector

correct

- ○ which coordinate is being varied to cause a change in the basis vector
- ○ which basis vector's change is being considered

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 3 of 10 (Start time: 12:43:20 PM, 25 January 2023)

The covariant differentiation of a tensor with constant components is zero.

Select the correct option

- False
- True

Question # 4 of 10 (Start time: 12:43:56 PM, 25 January 2023)

In generalized curvilinear coordinates the divergence of vector point function $A^i = A^i(x^i)$ is defined as_____.

Select the correct option

Relo

- $div(A^i) = A^i_{,i} + \Gamma^i_{ik}$
- $div(A^i) = A_{i,i} + \Gamma^i_{ik} A^k$
- $div(A^i) = A^i_{,i} + \Gamma^k_{ii} A^k$
- $div(A^i) = A^i_{,i} + \Gamma^i_{ik} A^k$

MTH623 - Tensor Analysis and its Applications (Quiz # 3) **Quiz Start Time**

Question # 1 of 10 (Start time: 12:42:19 PM, 25 January 2023)

The ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____

Select the correct option

- none of these
- quadratic transformations
- non-linear transformations
- linear transformations

Question # 7 of 10 (Start time: 12:47:06 PM, 25 January 2023)

The Christoffel symbols of first kind Γ_{ijk} are symmetric in_____.

Select the correct option

○	first two indices
○	second and third indices
○	first and third indices
○	all indices

correct

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 5 of 10 (Start time: 12:45:22 PM, 25 January 2023)

The Christoffel symbols of second kind are symmetric in_____.

Select the correct option

- all indices
- lower indices

correct

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 9 of 10 (start time: 12:48:57 PM, 25 January 2023)

The Christoffel symbols of first kind Γ_{ijk} are defined by _____

Select the correct option

- $\Gamma_{jki} = \frac{1}{2}(-g_{jk,i} + g_{ki,j} + g_{ki,j})$
- $\Gamma_{ijk} = \frac{1}{2}(g_{jk,i} + g_{ki,j} + g_{ki,j})$
- $\Gamma_{ij} = \frac{1}{2}(g_{jk,i} + g_{ki,j} + g_{ki,j})$
- $\Gamma_{ijk} = \frac{1}{2}(-g_{jk,i} + g_{ki,j} + g_{ki,j})$

correct

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 10 of 10 (start time: 12:50:44 PM, 25 January 2023)

The covariant differentiation of a scalar is equal to _____.

Select the correct option

- scalar field itself
- partial derivative of scalar field

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 1 of 10 (Start time: 01:19:32 PM, 25 January 2023)

The covariant differentiation of metric tensor is _____

Select the correct option

- zero in all coordinate systems
- zero in rectangular Cartesian coordinates only

Question # 2 of 10 (Start time: 01:19:40 PM, 25 January 2023)

The ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____

Select the correct option

- (●) linear transformations
- ( ) quadratic transformations
- ( ) non-linear transformations
- ( ) none of these

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 8 of 10 (Start time: 01:20:57 PM, 25 January 2023)

In curvilinear coordinates the derivative of a vector also includes_____

Select the correct option

- (●) the derivatives of basis vectors
- ( ) the derivatives of unit vectors

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 10 of 10 (Start time: 01:21:17 PM, 25 January 2023)

The Christoffel symbols of first and second kind are equal in any coordinate system.

Select the correct option

- (●) True
- ( ) False

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 2 of 10 (Start time: 12:42:53 PM, 25 January 2023)

In the notation Γ_{ij}^{k} of Christoffel symbol of second, the symbol k tells_____

Select the correct option

- ○ none of these
- ○ which basis vector points in the direction of this component of the derivative vector ✓
- ○ which coordinate is being varied to cause a change in the basis vector
- ○ which basis vector's change is being considered

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 3 of 10 (start time: 12:43:20 PM, 25 January 2023)

The covariant differentiation of a tensor with constant components is zero.

Select the correct option

- False
- True

Question # 4 of 10 (Start time: 12:43:56 PM, 25 January 2023)

In generalized curvilinear coordinates the divergence of vector point function $A^i = A^i(x^i)$ is defined as_.

Select the correct option

- $div(A^i) = A^i_{,i} + \Gamma^i_{ik}$
- $div(A^i) = A_{i,i} + \Gamma^i_{ik} A^k$
- $div(A^i) = A^i_{,i} + \Gamma^k_{ii} A^k$
- $div(A^i) = A^i_{,i} + \Gamma^i_{ik} A^k$

Question # 1 of 10 (Start time: 12:42:19 PM, 25 January 2023)

The ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____.

Select the correct option

- none of these
- quadratic transformations
- non-linear transformations
- linear transformations

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 7 of 10 (start time: 12:47:06 PM, 25 January 2023)

The Christoffel symbols of first kind Γ_{ijk} are symmetric in_____

Select the correct option

- ○ first two indices ✓
- ○ second and third indices
- ○ first and third indices
- ○ all indices

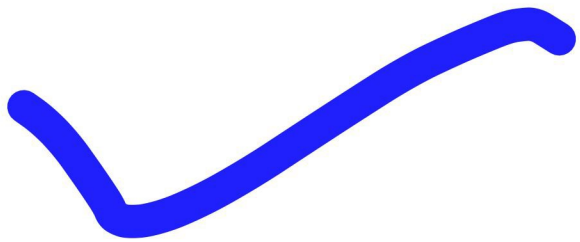
MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 8 of 10 (Start time: 12:47:54 PM, 25 January 2023)

The covariant derivative and partial derivatives coincide when _____ are constant.

Select the correct option

- ○ all components of metric tensor
- ○ Christoffel symbols



MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 5 of 10 (Start time: 12:45:22 PM, 25 January 2023

The Christoffel symbols of second kind are symmetric in__

Select the correct option

- all indices
- lower indices

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 6 of 10 (Start time: 12:46:25 PM, 25 January 2023)

The Christoffel symbols of first and second kind are equal in any coordinate system.

Select the correct option

- False
- True

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 9 of 10 (start time: 12:48:57 PM, 25 January 2023)

The Christoffel symbols of first kind Γ_{ijk} are defined by _____

Select the correct option

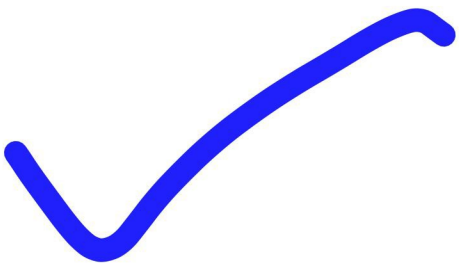
- $\Gamma_{jki} = \frac{1}{2}(-g_{jk,i} + g_{ki,j} + g_{ki,j})$
- $\Gamma_{ijk} = \frac{1}{2}(g_{jk,i} + g_{ki,j} + g_{ki,j})$
- $\Gamma_{ij} = \frac{1}{2}(g_{jk,i} + g_{ki,j} + g_{ki,j})$
- $\Gamma_{ijk} = \frac{1}{2}(-g_{jk,i} + g_{ki,j} + g_{ki,j})$ ✓

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 10 of 10 (Start time: 12:50:44 PM, 25 January 2023)

The covariant differentiation of a scalar is equal to _____.

Select the correct option

- scalar field itself 
- partial derivative of scalar field

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 1 of 10 (Start time: 01:19:32 PM, 25 January 2023)

The covariant differentiation of metric tensor is _____

Select the correct option

- zero in all coordinate systems
- zero in rectangular Cartesian coordinates only

Question # 2 of 10 (Start time: 01:19:40 PM, 25 January 2023)

The ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____

Select the correct option

- (●) linear transformations
- ( ) quadratic transformations
- ( ) non-linear transformations
- ( ) none of these

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 3 of 10 (Start time: 01:19:46 PM, 25 January 2023)

The Christoffel symbols of first kind satisfy the transformation law of a third rank tensor.

Select the correct option

- (●) False
- ( ) True

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 8 of 10 (Start time: 01:20:57 PM, 25 January 2023)

In curvilinear coordinates the derivative of a vector also includes_____

Select the correct option

- (●) the derivatives of basis vectors
- ( ) the derivatives of unit vectors

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 7 of 10 (Start time: 01:20:51 PM, 26 January 2023)

Christoffel symbols of first and second kind include derivatives of _____

Select the correct option

- (●) basis vectors
- ( ) unit vectors
- ( ) metric tensor
- ( ) normal vectors

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 6 of 10 (Start time: 01:20:31 PM, 25 January 2023)

The Christoffel symbols of first kind Γ_{ijk} are defined by _____

Select the correct option

○	$\Gamma_{jki} = \frac{1}{2}(-g_{jk,i} + g_{ki,j} + g_{ki,j})$
○	$\Gamma_{ij} = \frac{1}{2}(g_{jk,i} + g_{ki,j} + g_{ki,j})$
◉	$\Gamma_{ijk} = \frac{1}{2}(-g_{jk,i} + g_{ki,j} + g_{ki,j})$
○	$\Gamma_{ijk} = \frac{1}{2}(g_{jk,i} + g_{ki,j} + g_{ki,j})$

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 10 of 10 (Start time: 01:21:17 PM, 25 January 2023)

The Christoffel symbols of first and second kind are equal in any coordinate system.

Select the correct option

- True
- False

Rank of a tensor derivative:

The derivative of a zero order tensor becomes a first order tensor.
Generally, the derivative of an m- order tensor forms an $m+1$ order tensor.

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Question # 5 of 10 (Start time: 03:06:34 PM, 25 January 2023)

In any coordinate system x^i, the covariant derivative with respect to x^k of a covariant vector T_i is defined as_____.

Select the correct option

- $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^t_{ik} T_t$
- $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^i_{ik} T_t$
- $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^t_{ik}$
- $T_{i;k} = \frac{\partial T_i}{\partial x^k} + \Gamma^t_{ik} T_t$

The covariant derivative of an arbitrary tensor is a tensor.

Select the correct option

○	False
○	True

estion # 2 of 10 (Start time: 12:14:14 PM, 25 January 2023)

The Christoffel symbols of first kind satisfy the transformation law of a third rank tensor.

lect the correct option

- True
- False

Question # 3 of 10 (Start time: 12:16:16 PM, 25 January 2023)

In generalized curvilinear coordinates the gradient of scalar function $\phi = \phi(x^i)$ is defined as __

Select the correct option

○	$\nabla\phi = \frac{\partial\phi}{\partial x^i}$
○	$\nabla\phi = \partial\phi\hat{e}_i$
○	$\nabla\phi = \frac{\partial\phi}{\partial x^i}\hat{e}_i$
○	$\nabla\phi = \frac{\partial x^i}{\partial\phi}$

correct

Cl

For a diagonal metric which of the following is true?

Select the correct option

○	$\Gamma_{iii} = \frac{1}{2} g_{ij,j}$
○	$\Gamma_{iii} = \frac{1}{2} g_{ii,i}$

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 7 of 10 (Start time: 08:32:40 PM, 24 January 2023)

The Christoffel symbols of first kind satisfy the transformation law of a third rank tensor.

Select the correct option

○	False correct
○	True

Question # 4 of 10 (Start time: 08:26:32 PM, 24 January 2023)

Christoffel symbols of first and second kind include derivatives of _____.

Select the correct option

○	unit vectors
○	basis vectors
○	metric tensor
○	normal vectors

correct

Question # 2 of 10 (Start time: 08:23:11 PM, 24 January 2023)

The covariant differentiation of a tensor with constant components is zero.

Select the correct option

○	True
○	False

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 1 of 10 (Start time: 08:21:12 PM, 24 January 2023)

In Cartesian coordinate system all the Christoffel symbols are _____

Select the correct option

- zero
- vanish
- equal
- all options are true

Question # 2 of 10 (Start time: 08:23:11 PM, 24 January 2023)

The covariant differentiation of a tensor with constant components is zero.

Select the correct option

○	True
○	False

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 1 of 10 (Start time: 08:21:12 PM, 24 January 2023)

In Cartesian coordinate system all the Christoffel symbols are _____

Select the correct option

- zero
- vanish
- equal
- all options are true

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 1 of 10 (Start time: 08:21:12 PM, 24 January 2023)

In Cartesian coordinate system all the Christoffel symbols are _____

Select the correct option

○	zero
○	vanish
○	equal
○	all options are true

Question # 2 of 10 (Start time: 08:23:11 PM, 24 January 2023)

The covariant differentiation of a tensor with constant components is zero.

Select the correct option

○	True
○	False

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 3 of 10 (Start time: 08:24:55 PM, 24 January 2023)

In the notation Γ_{ij}^{k} of Christoffel symbol of second, the symbol k tells_____

Select the correct option

○	none of these
○	which coordinate is being varied to cause a change in the basis vector
○	which basis vector's change is being considered
○	which basis vector points in the direction of this component of the derivative v

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 3 of 10 (Start time: 08:24:55 PM, 24 January 2023)

In the notation Γ_{ij}^{k} of Christoffel symbol of second, the symbol k tells_____

Select the correct option

○	none of these
○	which coordinate is being varied to cause a change in the basis vector
○	which basis vector's change is being considered
○	which basis vector points in the direction of this component of the derivative v

correct

Question # 4 of 10 (Start time: 08:26:32 PM, 24 January 2023)

Christoffel symbols of first and second kind include derivatives of _____

Select the correct option

- ○ unit vectors
- ○ basis vectors
- ○ metric tensor
- ○ normal vectors

correct

Question # 5 of 10 (Start time: 08:28:43 PM, 24 January 2023)

The covariant differentiation of metric tensor is _____.

Select the correct option

○	zero in all coordinate systems
○	zero in rectangular Cartesian coordinates only



MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 7 of 10 (Start time: 08:32:40 PM, 24 January 2023)

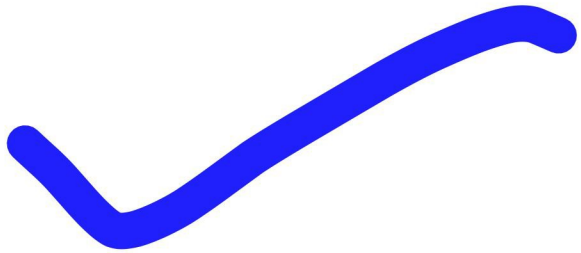
The Christoffel symbols of first kind satisfy the transformation law of a third rank tensor.

Select the correct option

- ○ False
- ○ True

The Christoffel symbols of first kind satisfy the transformation law of a third rank tensor.

Select the correct option

- ○ False 
- ○ True

The covariant differentiation of metric tensor is _____.

Select the correct option

- zero in all coordinate systems
- zero in rectangular Cartesian coordinates only

MTH623 - Tensor Analysis and its Applications (Quiz # 3)

Question # 9 of 10 (Start time: 04:31:08 PM, 25 January 2023)

In generalized curvilinear coordinates the gradient of scalar function $\phi = \phi(x^i)$ is defined as _

Select the correct option

- $\nabla\phi = \frac{\partial\phi}{\partial x^i}$
- $\nabla\phi = \frac{\partial x^i}{\partial\phi}$
- $\nabla\phi = \partial\phi\hat{e_i}$
- $\nabla\phi = \frac{\partial\phi}{\partial x^i}\hat{e_i}$

Question # 1 of 10 (start time: 06:03:46 PM, 25 January 2023)

For a diagonal metric which of the following is true?

Select the correct option

- $\Gamma_{iii} = \frac{1}{2} g_{ii,i}$
- $\Gamma_{iii} = \frac{1}{2} g_{ij,j}$

Question # 3 of 10 (Start time: 01:40:35 PM, 25 January 2023)

In any coordinate system x^i, the covariant derivative with respect to x^k of a covariant vector T_i is defined as_____

Select the correct option

- $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^i_{ik} T_l$
- $T_{i;k} = \frac{\partial T_i}{\partial x^k} + \Gamma^l_{ik} T_l$
- $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^l_{ik} T_l$
- $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^l_{ik}$

Click to Save Answer & N

uestion # 5 of 10 (Start time: 06:19:22 PM, 25 January 2023)

In any coordinate system x^i, the covariant derivative with respect to x^k of a covariant vector T_i is defined as_____

lect the correct option

- $T_{i;k} = \frac{\partial T_i}{\partial x^k} + \Gamma^t_{ik} T_t$
- $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^i_{ik} T_t$
- $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^t_{ik}$
- $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^t_{ik} T_t$

Click to Save Answer

Question # 10 of 10 (start time: 06:10:30 PM, 25 January 2023) **Total Marks:** 1

The covariant derivative of a tensor of order n is a tensor of order _____.

Select the correct option

Reload Math Equations

○	$n+1$
○	n
○	$2n$
○	$n-1$

Click to Save Answer & Move to Next Question

Which of the following is true for basis vectors in general curvilinear coordinates systems?

Select the correct option

Reload Math Equations

- $\frac{\partial \vec{e}_i}{\partial x^j} = \vec{e}_k$
- $\frac{\partial \vec{e}_i}{\partial x^j} = \Gamma^k_{ij} \vec{e}_k$
- $\frac{\partial \vec{e}_i}{\partial x^j} = 0$
- $\frac{\partial \vec{e}_i}{\partial x^j} = \Gamma^k_{ij}$

Click to Save Answer & Move to Next Question